

Signal sampling, noise and other parameters

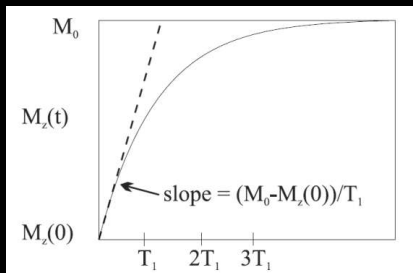
Jaladhar Neelavalli, PhD.,

Recap

$$\frac{d\vec{M}}{dt} = \gamma \cdot \vec{M} \times B + \frac{M_0 - M_z}{T_1} - \frac{M_{xy}}{T_2}$$

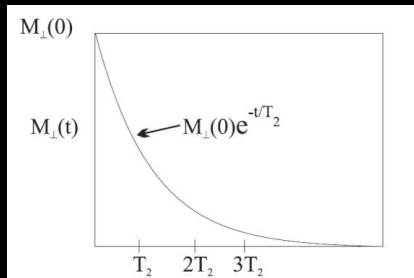
Bloch equation

$$M_z(t) = M_z(0) \cdot e^{-\frac{t}{T_1}} + M_0 \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$$

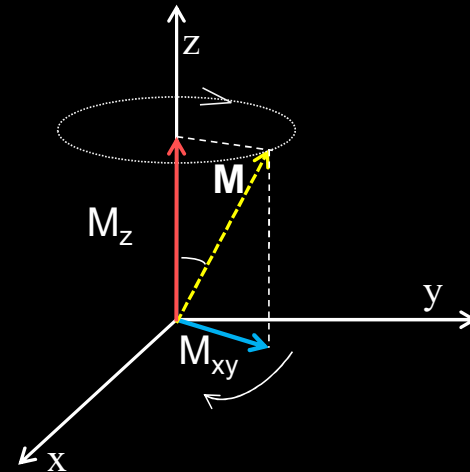
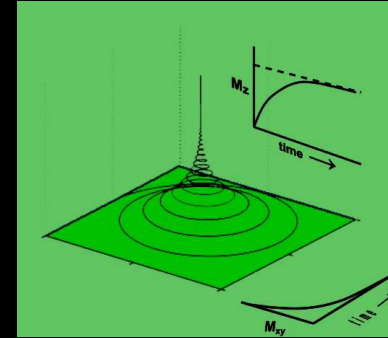


T1 recovery –
Spin lattice relaxation

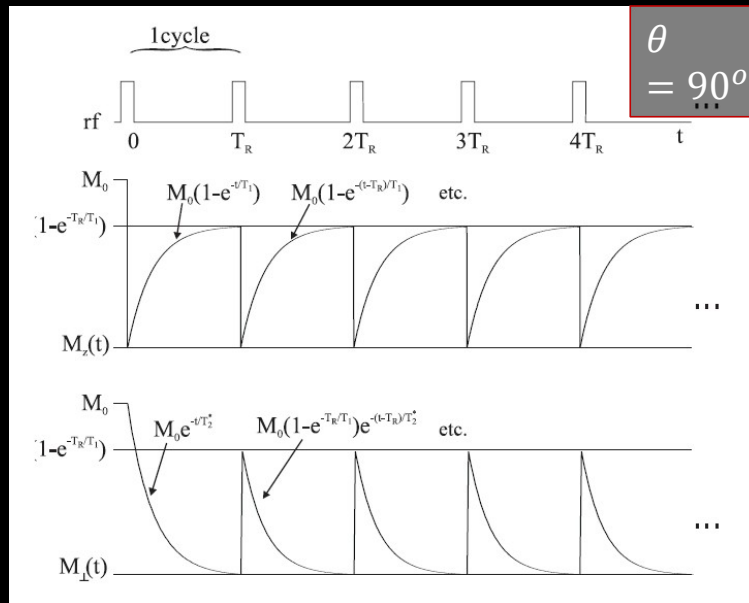
$$M_{xy}(t) = M_{xy}(0) \cdot e^{-\frac{t}{T_2}}$$



T2 decay –
Spin-spin relaxation



Recap



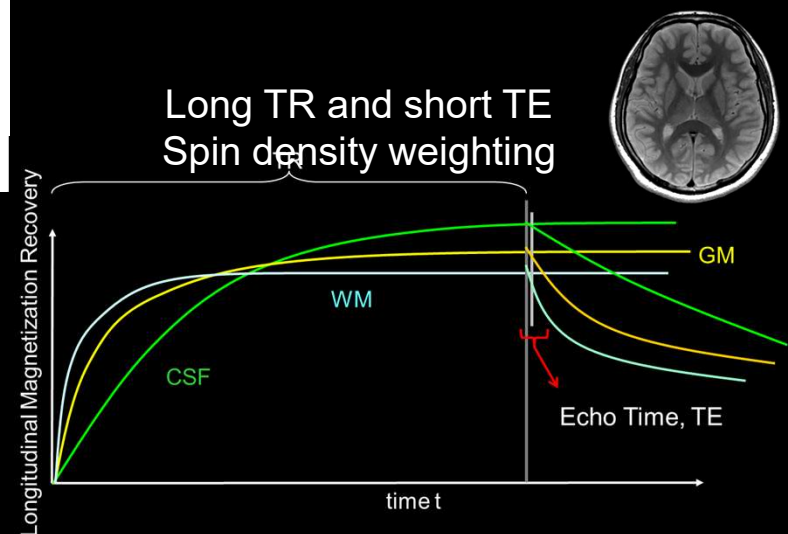
Imaging experiment- multipulse – steady state signal

Long and short here are terms relative to T_1 and T_2 's of the tissues...
 $TR \gg T_1, TE \ll T_2$

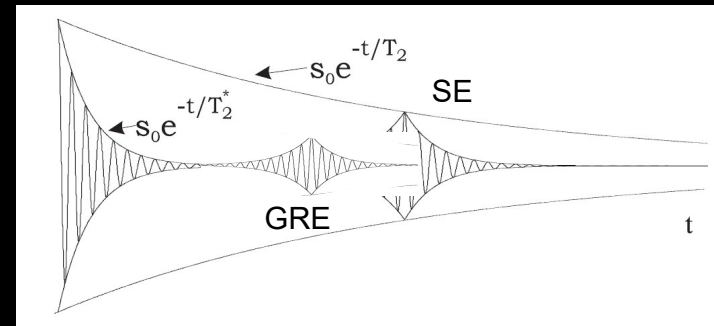
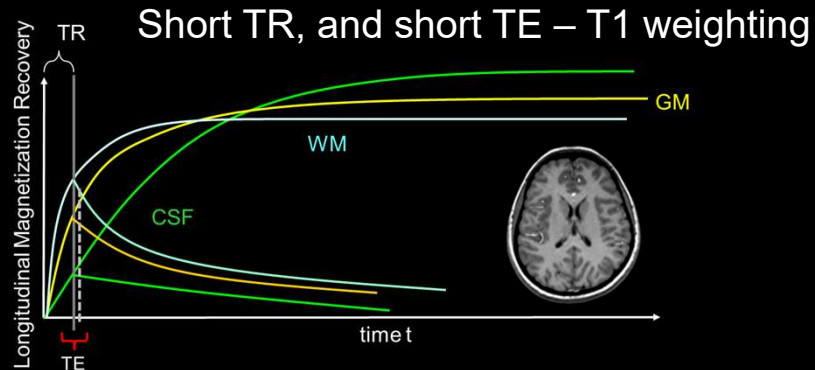
Steady state reached by the second rf pulse

Steady state meaning... $M_z(TR)$ and $M_{xy}(TE)$ are constant from one TR to the next

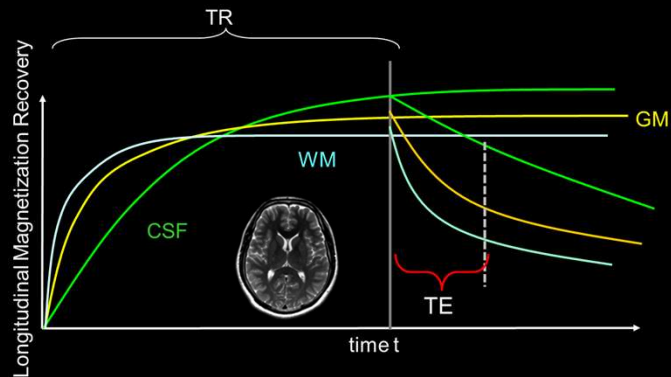
$$S = M_{xy}(TE) = M_0 \cdot \left(1 - e^{-\frac{TR}{T_1}}\right) \cdot e^{-\frac{TE}{T_2}}$$



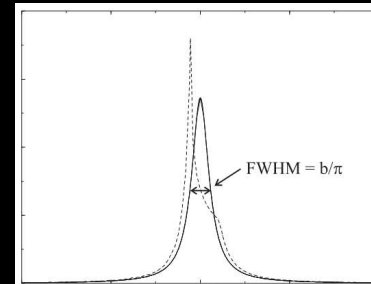
Recap



Long TR, and Long TE – T2 weighting

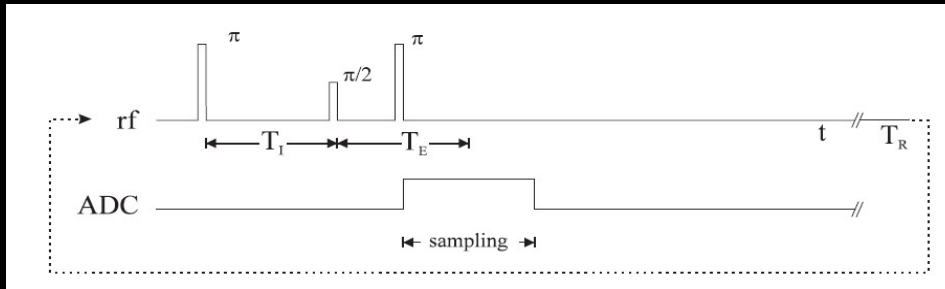


$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_2'}$$

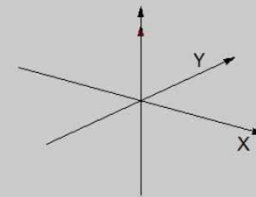


$$T_2' \approx \frac{1}{|\gamma \cdot \Delta B|}$$

Recap

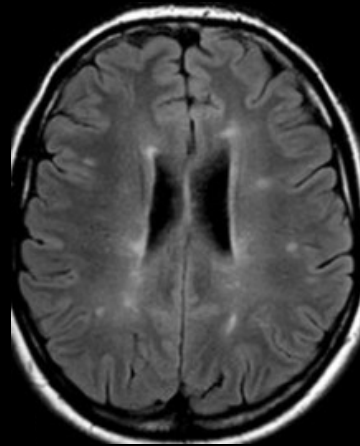


Inversion recovery

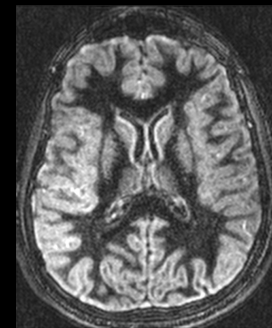
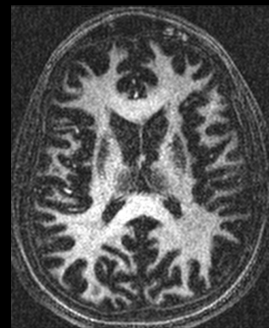
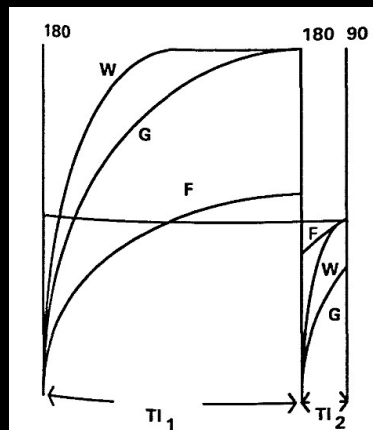
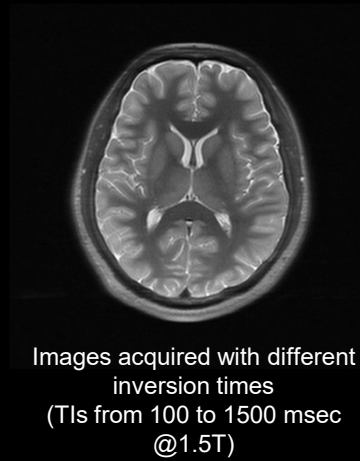
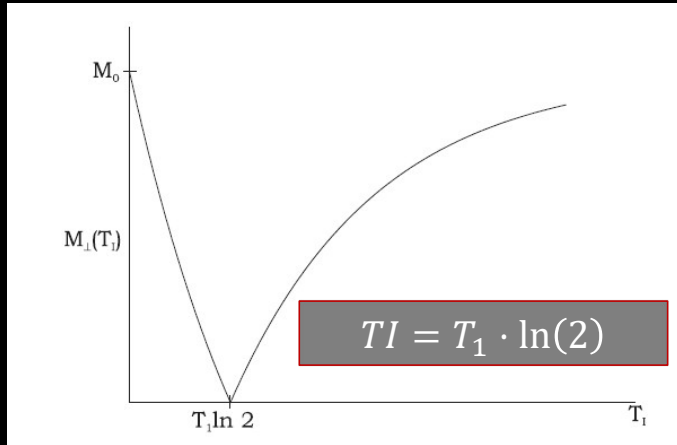


FLAIR-

Fluid
attenuated
Inversion
recovery

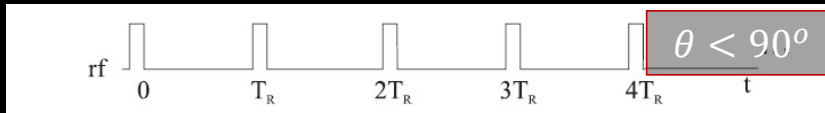


Recap

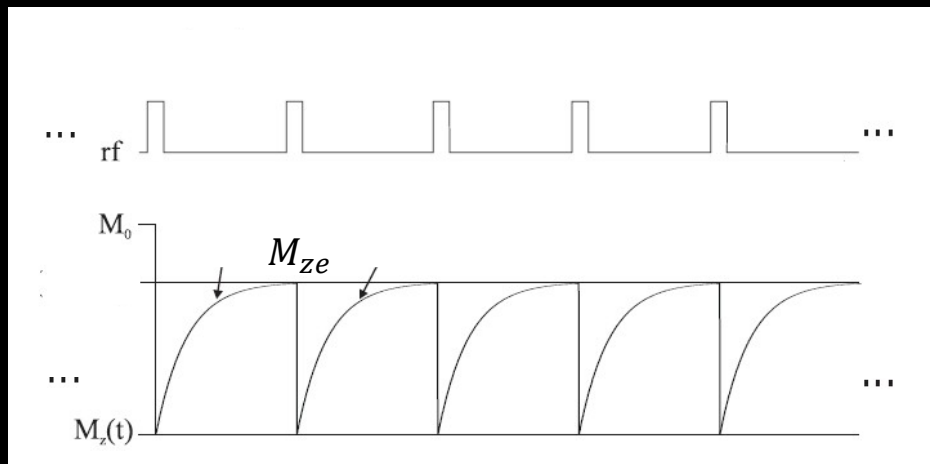


Double inversion recovery

Recap



SPGR – spoiled gradient echo
FLASH – Fast low angle shot



$$S = \frac{M_0 \left(1 - e^{-\frac{TR}{T_1}}\right) \cdot \sin(\theta) \cdot e^{-\frac{TE}{T_2^*}}}{\left(1 - \cos(\theta) \cdot e^{-\frac{TR}{T_1}}\right)}$$

Now, this is dependent not only on TR and TE, but also on the tipping angle (flip angle)

FLASH

$$S = \frac{M_0 \left(1 - e^{-\frac{TR}{T_1}}\right) \cdot \sin(\theta) \cdot e^{-\frac{TE}{T_2^*}}}{\left(1 - \cos(\theta) \cdot e^{-\frac{TR}{T_1}}\right)}$$

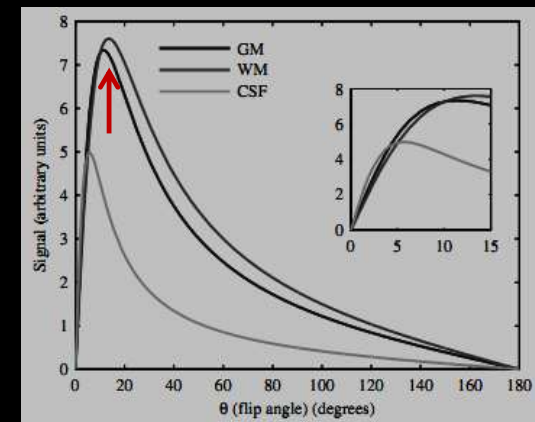
Optimizing signal as a function of flip angle

$$\frac{dS}{d\theta} = 0 \quad \longrightarrow \quad \cos(\theta_E) = e^{-\frac{TR}{T_1}}$$

Ernst Angle

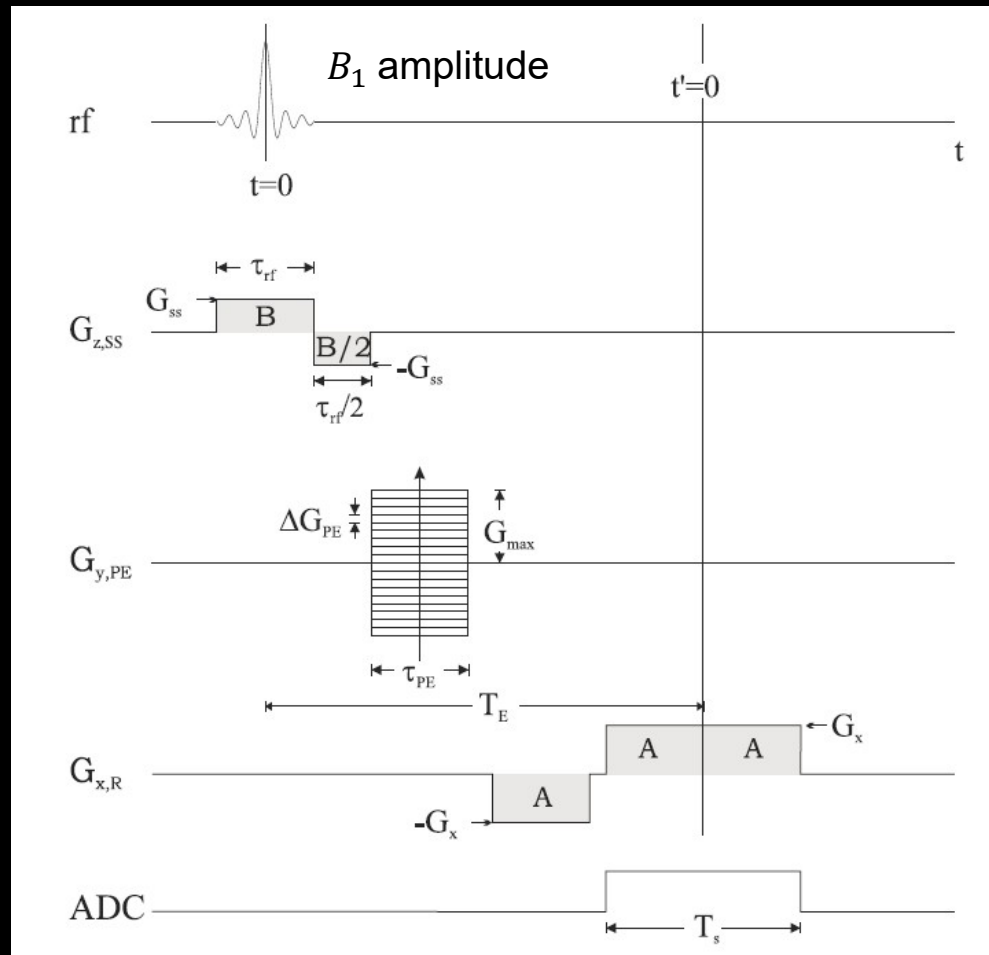
Simplifying this

$$\theta_E \simeq \sqrt{\frac{2 \cdot TR}{T_1}}$$



So, what are the amplitudes of all these sequence variables and how do they relate to the object you want to image?

- Field of view we are imaging in read and phase-encode directions (FOV_x , FOV_y)
- Spatial resolution we want in read and phase-encode directions (Δx , Δy)
..... $N_x = FOV_x / \Delta x$
- TR
- TE
- Flip angle

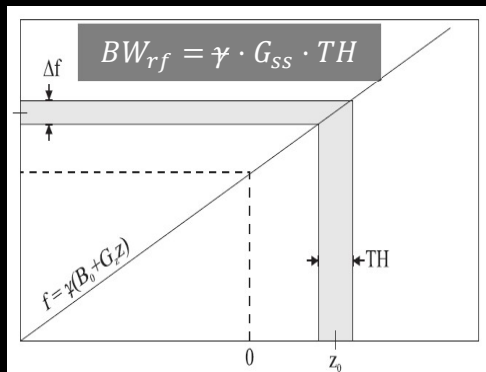


- B_1 – "amplitude of the rf pulse"
- τ_{rf} – duration of the rf pulse
- G_{ss} – duration of slice select grad.
- ΔG_{pe} – step size of phase encode G
- τ_{pe} – duration of the phase encode G
- N_{pe} (or G_{pe-max}) – Number of phase encodes
- G_{read} – read grad. amplitude
- T_s – total sampling duration
- Dwell time Δt during sampling

Imaging Experiment

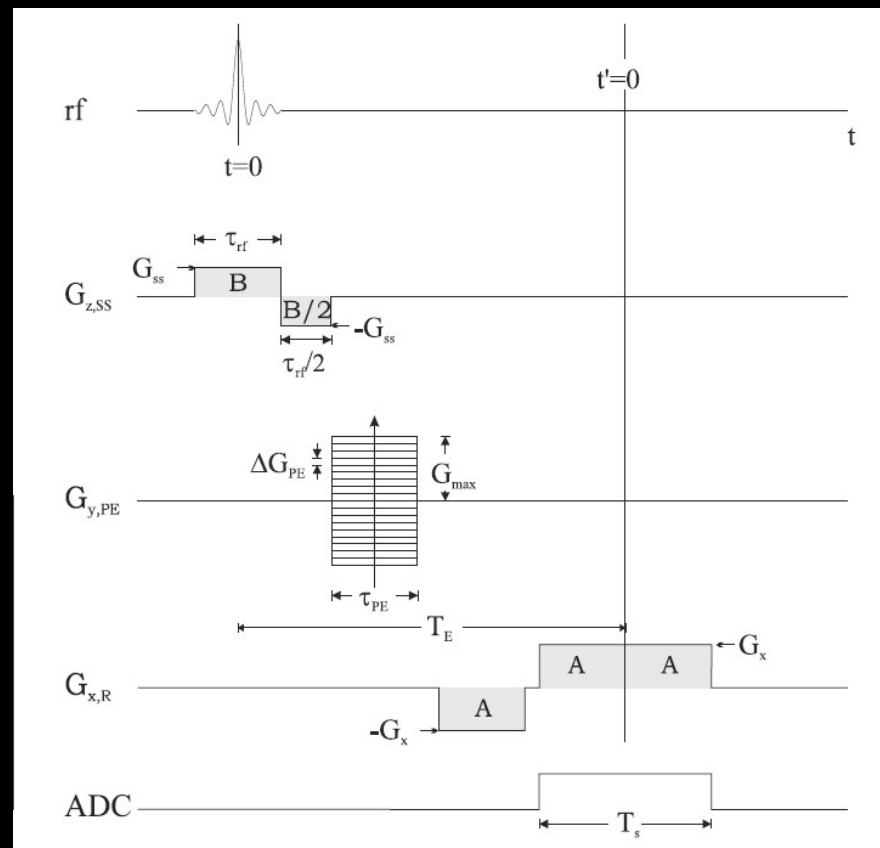
Where do you start?

Slice selection



Experimental parameters

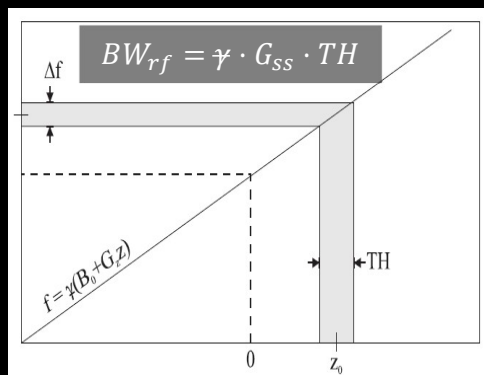
slice thickness- TH ,
 τ_{rf} ,
 rf pulse bandwidth BW_{rf} ,
 slice selection gradient G_{ss}



Imaging Experiment

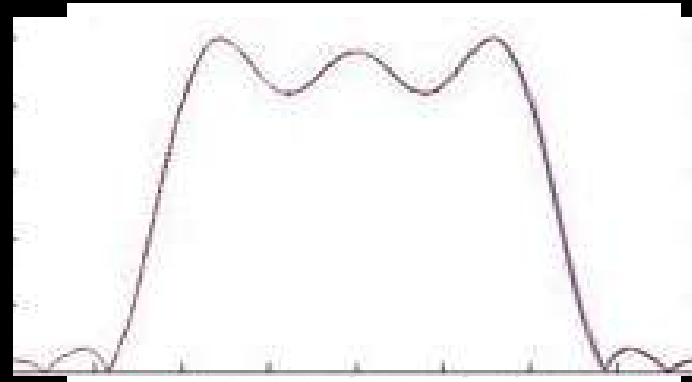
Where do you start?

Slice selection

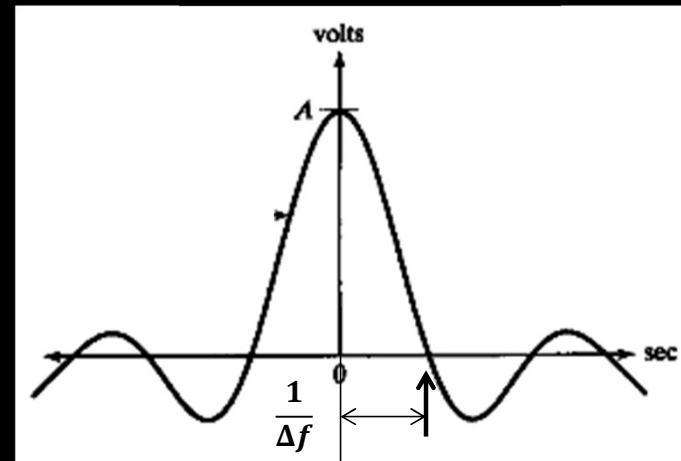


$$\Delta f = BW_{rf}$$

$$\tau_{rf} = n_{zc} \cdot \left(\frac{1}{BW_{rf}} \right)$$



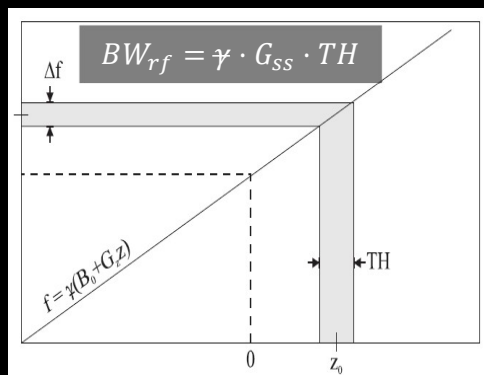
$$BW_{rf} = \Delta f = \gamma \cdot G_{ss} \cdot TH$$



Imaging Experiment

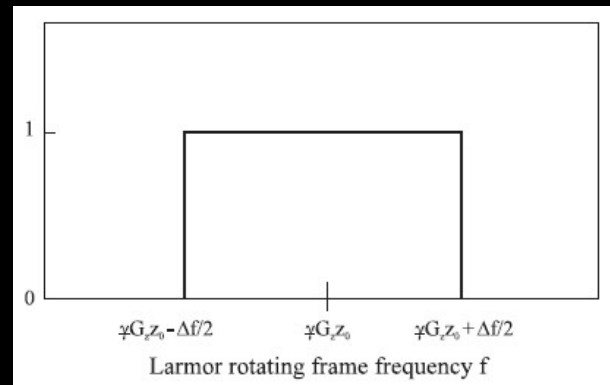
Where do you start?

Slice selection

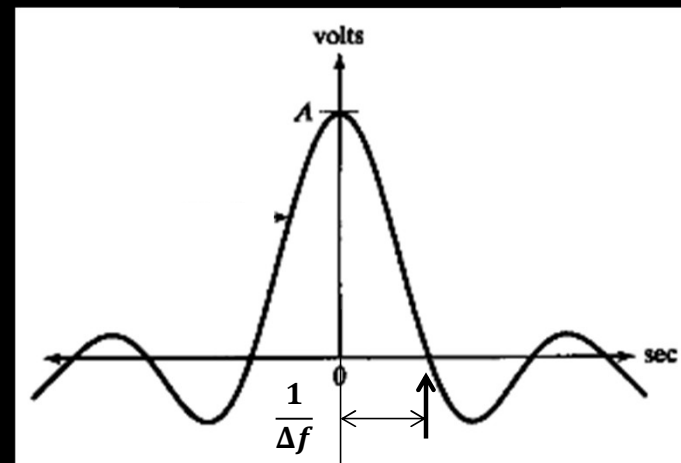


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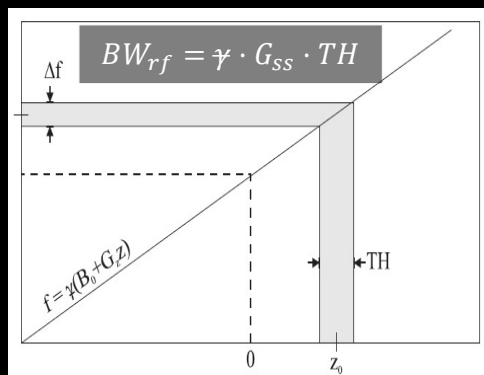


$$BW_{rf} = \Delta f = \gamma \cdot G_{ss} \cdot TH$$



Imaging Experiment

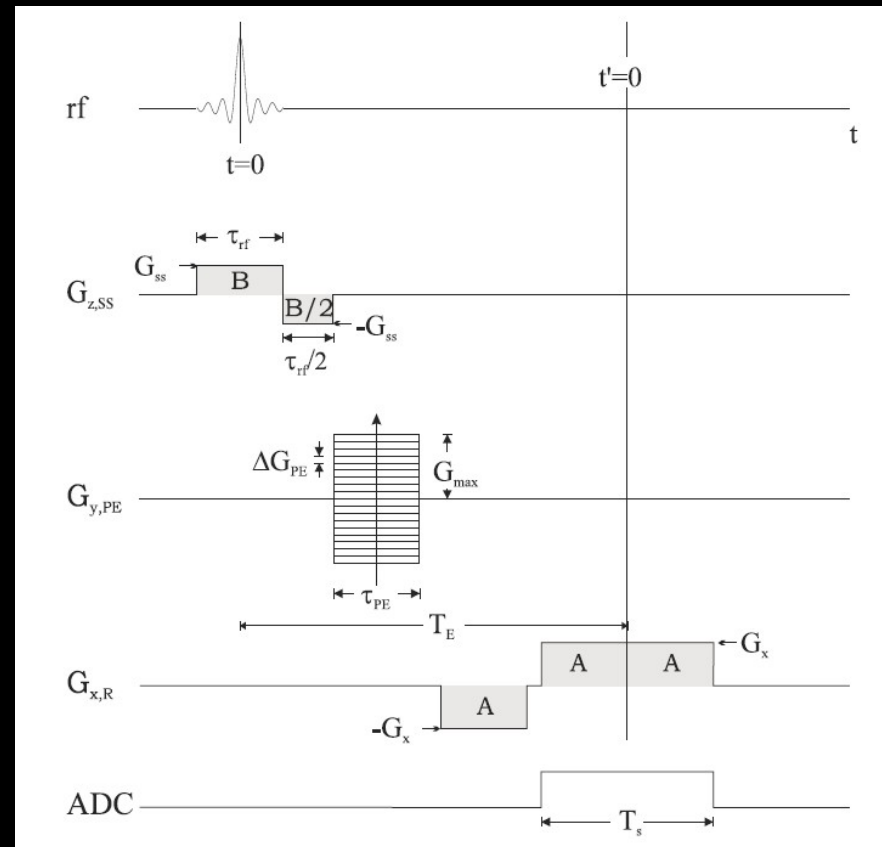
Slice selection



$$\Delta f = BW_{rf}$$

$$\tau_{rf} = n_{zc} \cdot \left(\frac{1}{\Delta f} \right)$$

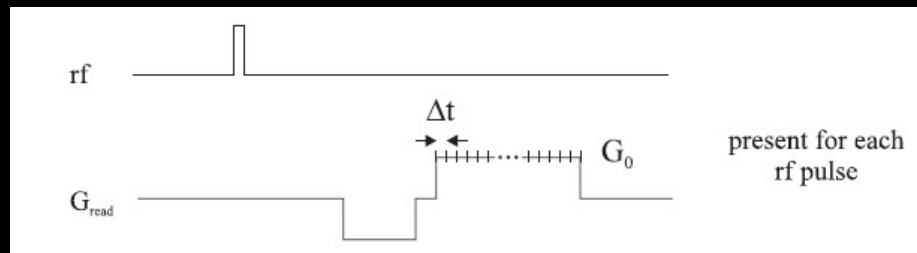
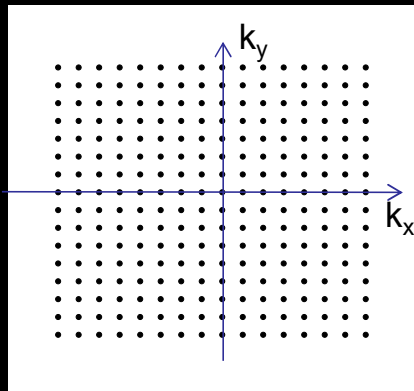
First decide, TH , then rf bandwidth and n_{zc} , which will in-turn determines τ_{rf} and G_{ss}



Imaging equation & k-space

3D imaging signal equation

$$s(k_x, k_y, k_z) = \iiint dx \cdot dy \cdot dz \cdot \rho(x, y, z) \cdot e^{-i2\pi (k_x x + k_y y + k_z z)}$$



$$k_x = \gamma \cdot G_x \cdot t = \gamma \cdot G_x \cdot \Delta t$$

$$\Delta k_x = \gamma \cdot G_x \cdot \Delta t$$

Sampling the signal in k-space

Imaging equation & k-space

$$\Delta k_x = \gamma \cdot G_x \cdot \Delta t$$

Sampled signal

$$s(p' \Delta k_x, q' \Delta k_y, r' \Delta k_z) = \sum_{p,q,r} \rho(p \Delta x, q \Delta y, r \Delta z) \cdot e^{-i2\pi \left(\frac{p \cdot p'}{N_x} + \frac{q \cdot q'}{N_y} + \frac{r \cdot r'}{N_z} \right)}$$

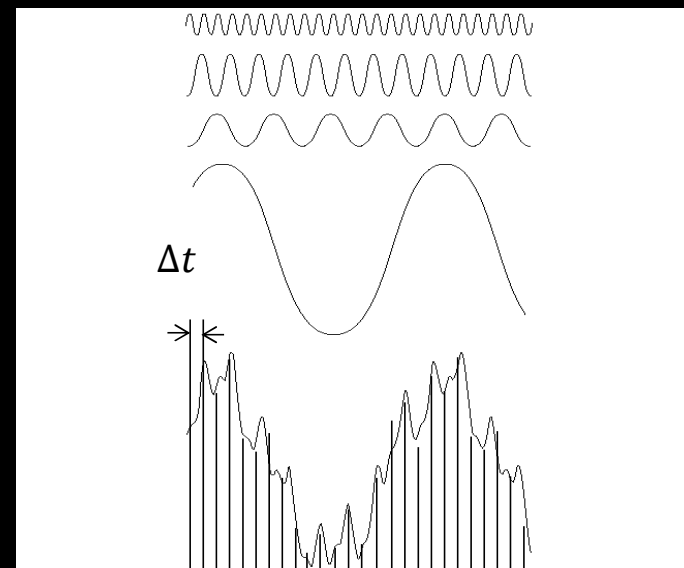
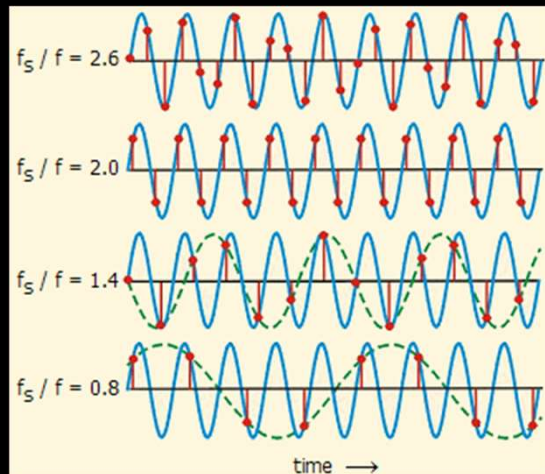
Reconstructed image

$$\rho(p \Delta x, q \Delta y, r \Delta z) = \frac{1}{N_x N_y N_z} \sum_{p',q',r'} s(p' \Delta k_x, q' \Delta k_y, r' \Delta k_z) \cdot e^{i2\pi \left(\frac{p \cdot p'}{N_x} + \frac{q \cdot q'}{N_y} + \frac{r \cdot r'}{N_z} \right)}$$

So, what should the sampling rate be in k-space?

Nyquist relation

If we have a continuous signal like this...



f_{max}

$$2 \cdot f_{max} = \frac{1}{\Delta t}$$

To digitally reconstruct this signal reliably, the rate at which you sample this has to be at least twice the largest frequency in the signal

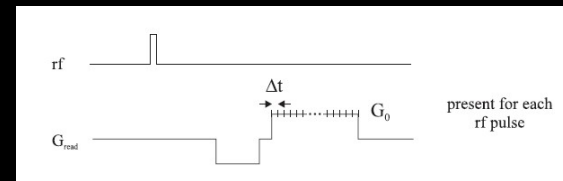
Nyquist relation

As a consequence of this condition, we have the following sampling relation in MRI

$$\gamma G_x L_x = \frac{1}{\Delta t}$$

$$\gamma G_x \Delta t = \frac{1}{L_x}$$

$$\Delta k_x = \frac{1}{L_x}$$



Where L_x is the field of view of our imaging experiment in the read-out direction

Nyquist relation

In phase encode direction,

$$\Delta k_y = \frac{1}{L_y}$$

$$\gamma \Delta G_y \tau = \frac{1}{L_y}$$

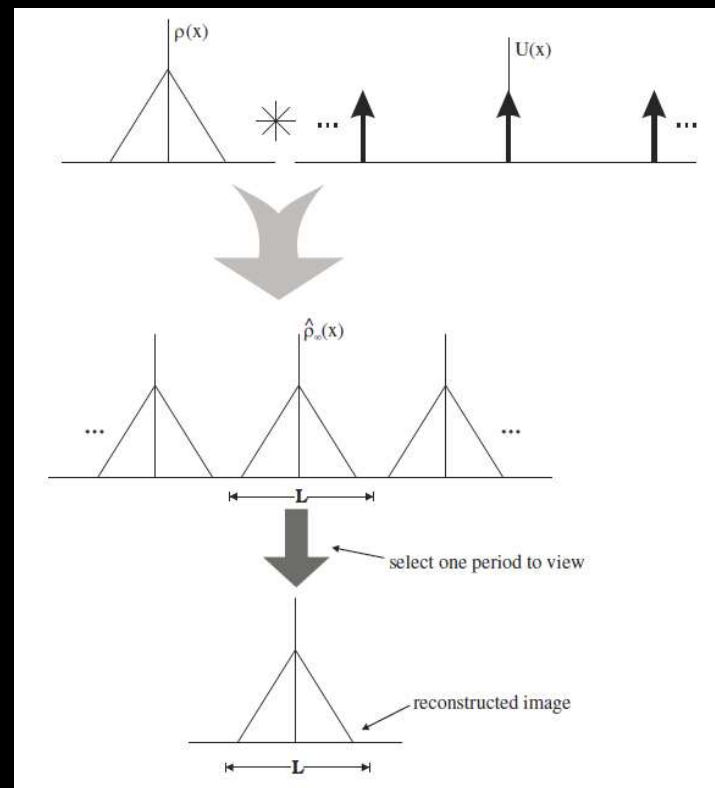
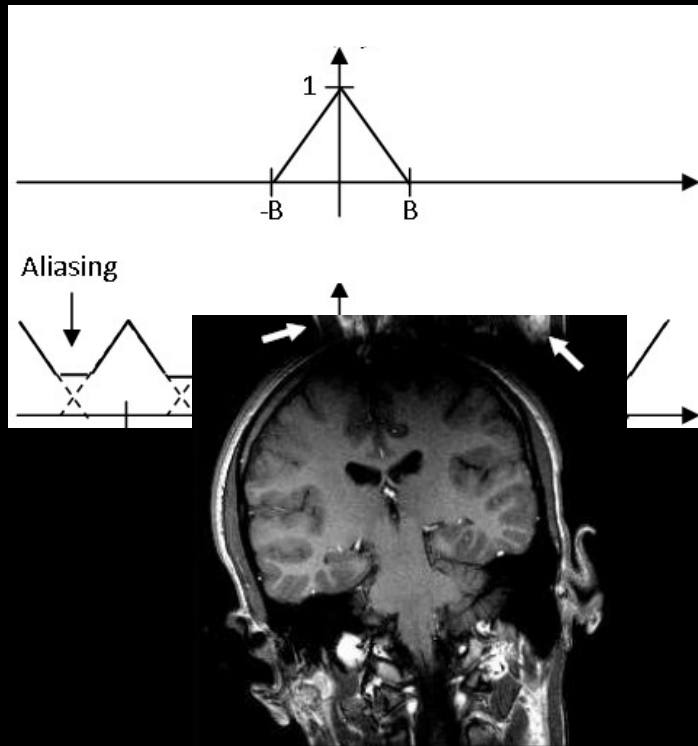
$$\Delta k_y = \frac{1}{L_y}$$

Where L is the field of view of our imaging experiment in the phase encoding direction

Nyquist relation

Such a sampling also avoids aliasing as long as FOV > dimensions of the object

$$L = \frac{1}{\Delta k}$$



Imaging Experiment

Where do you start?

First L – FOV - L_x and L_y

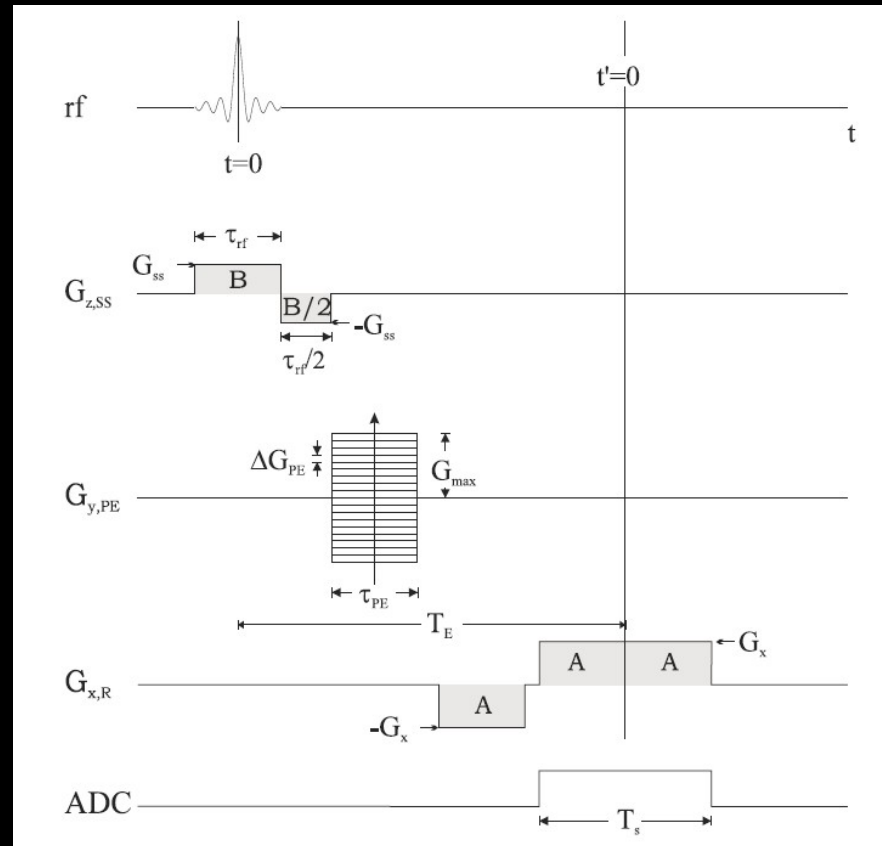
$$\Delta k_x = \frac{1}{L_x} = \frac{1}{N_x \cdot \Delta x}$$

$$\Delta k_x = \gamma \cdot G_x \cdot \Delta t$$

So, Δt and G_x can then be decided

$$\Delta k_y = \frac{1}{L_y} = \gamma \Delta G_y \tau = \frac{1}{N_y \cdot \Delta y}$$

So, ΔG_y and τ can then be decided



Imaging Experiment

How long should we sample?

$$\Delta k_x = \frac{1}{L_x} = \frac{1}{N_x \cdot \Delta x}$$

$$\frac{1}{N_x \cdot \Delta x} = \gamma \cdot G_x \cdot \Delta t$$

$$\frac{1}{\gamma \cdot G_x \cdot \Delta x} = N_x \cdot \Delta t = T_s$$

So, once you fix Δx , N_x and G_x , then Δt and hence T_s gets determined

$\gamma \cdot G_x \cdot \Delta x$ Bandwidth per voxel

$\gamma \cdot G_x \cdot N_x \cdot \Delta x$ Total read bandwidth

